

PharmaSync

A clinic pharmacy workflow system for Controlled Drug (UKK) quota management, digital antibiotic (AMS) forms and AI-assisted antibiotic dose suggestions.

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1. About the project

PharmaSync is a web application built by the clinic's own Innovation Team. It replaces paper forms, phone confirmations, WhatsApp screenshots and hand-kept spreadsheets with one system shared by medical officers, the Family Medicine Specialist (FMS) and pharmacists. It has three main features: Controlled Drug (UKK) quota management, a digital antibiotic (AMS) form, and AI-assisted antibiotic dose suggestions.

Item	Detail
Innovation category	Technology innovation (web system written in TypeScript with a PostgreSQL database)
Project started	17 March 2026
In live use since	1 August 2026, as a trial in real clinic operations
Staff involved in the trial	4 Pharmacists, 1 Family Medicine Specialist, 1 Medical Officer
Running cost	About RM3.32 a day (domain RM13 a year, server RM100 a month)
Paper and toner saved	RM3,277.40 a year; RM2,064.40 net after the running cost

2. The problem

- Controlled Drug (UKK) approval from the FMS was given by phone, then screenshotted and posted to a WhatsApp group as evidence for the pharmacy. There was no official record and no status anyone could track.
- The annual UKK quota balance was worked out by hand in an Excel sheet at the pharmacy counter. The FMS could not see the current balance before agreeing to start a patient on a UKK.
- Controlled antibiotics needed a paper AMS form. On average 10 AMS forms a month arrived incomplete, and each time the patient was sent back to the doctor's room while the pharmacy waited.
- There was no central record to measure compliance with the National Antimicrobial Guideline (NAG) 2024, and doctors looked up antibiotic doses by hand from the guideline book.

3. Feature 1: Controlled Drug (UKK) quota and approval

This is the main feature of PharmaSync. It covers the whole path of a Controlled Drug, from the medical officer's request to the FMS decision to dispensing at the counter.

What it does

- The medical officer submits the UKK request directly in PharmaSync.
- The system shows the remaining annual quota before the request is sent, and blocks any request above the allocation.
- The quota is pooled at district level (PKDJB) and shared by all clinics in the district. It is set as a rate per FMS, multiplied by the number of FMS in the district, so nobody has to recalculate it by hand each year.
- Pesara patients (retired government employees) are exempt from the annual quota, and the system applies this automatically.
- The request goes straight to the FMS dashboard. The FMS sees the current quota balance and approves or rejects the request with a reason. The decision is recorded with the specialist's name and a timestamp. If the quota has run out by the time the FMS approves, the system shows a warning first.
- At the counter, the screening pharmacist logs in to PharmaSync and sees the approved request: patient name, IC or RN number, drug, dose and frequency, the approving FMS, the approval time and the quota balance.
- Each dispensing updates the quota drug patient list and the quota usage automatically. The patient's full medication history is still checked in PhIS.

What stays the same

The medical officer still discusses the patient with the FMS by phone or in person, and the FMS still countersigns the paper prescription. What changed is that the approval is now recorded in the system, and the quota balance is shown before the decision is made.

Before and after

	Before (manual)	After (PharmaSync)
Steps in the workflow	12	10
Rework cycles	1: pharmacy phones the medical officer to confirm the FMS discussion happened, and with which FMS, while the patient waits	None
Evidence of FMS approval	Screenshot in a WhatsApp group	Recorded in the system with name and timestamp
Quota balance	Excel sheet at the counter, calculated by hand	Calculated by the system, shown to the MO and the FMS
Time at the pharmacy counter	About 20 minutes	10 to 15 minutes

For each UKK patient that is 5 to 10 minutes less waiting, a reduction of 25% to 50%. The time runs from the patient taking a queue number at the pharmacy counter to the medication being dispensed.

4. Feature 2: Digital antibiotic (AMS) form

What it does

- The paper AMS form is replaced by a digital form that the medical officer fills in when prescribing. It carries a clinical checklist for each NAG 2024 pathway (pneumonia, acute otitis media, pharyngitis, rhinosinusitis, skin and soft tissue infection, urinary tract infection).
- The form cannot be submitted while any mandatory field is empty, so an incomplete form never reaches the pharmacy.
- Each controlled antibiotic prescription is checked against the Clinical Pathway of the National Antimicrobial Guideline (NAG) 2024 when it is written.
- The system recognises specialist antibiotics automatically. The FMS decision is recorded with the antibiotic code, and the approval status appears on screen for the pharmacist.
- The AMS record is created automatically, with no manual entry into an Excel list.

Before and after

	Before (manual)	After (PharmaSync)
Steps in the workflow	10	9
Rework cycles	2: patient sent back to the doctor to complete the AMS form; pharmacy phones the doctor for FMS approval and the antibiotic code	None
Incomplete AMS forms per month	10 on average	0
Dispensing time, correct dose	15 minutes	Under 7 minutes (about 53% less)
Dispensing time, dose needs correcting	Up to 25 minutes	Within 15 minutes (40% less)

Most of the time saved is at the prescription screening step, where the manual data entry used to happen. Compliance with NAG 2024 can now be monitored for every controlled antibiotic prescription. Before PharmaSync there was no central record to measure it.

5. Feature 3: AI-assisted antibiotic dose suggestion

What it does

- While the medical officer fills in the antibiotic form, the system suggests an antibiotic regimen from NAG 2024 based on the diagnosis, the clinical checklist, the patient's age, weight and allergy status.
- Children's doses are calculated by body weight (mg/kg) and capped at the maximum dose. Preferred and alternative regimens are shown, and a penicillin-class regimen is flagged when the patient has a penicillin allergy.
- A local AI model (Ollama, hosted on the clinic's own server in Malaysia) matches the diagnosis to dosing notes written by the pharmacists from NAG 2024, and shows that note word for word. The AI does not write or invent a dose.
- The suggestion goes into the prescription only when the doctor clicks "Use". It is never filled in automatically.
- If no NAG 2024 pathway matches the case, the system recommends referral to the specialist instead of suggesting a dose.
- Clinical data does not leave the country, and every AI use is written to the audit log.

Why it matters

Doctors no longer look up antibiotic doses by hand from the guideline book. The dose, the pathway check and the G6PD warning are all done while the prescription is written, before the patient reaches the pharmacy. Most of the time saved at the pharmacy screening step comes from this: the pharmacist checks a prescription that has already been screened.

6. Other features

Prescribing safety

- G6PD deficiency checker: warns the prescriber automatically when a drug carries a risk for a patient with G6PD deficiency.
- Paediatric dose calculator: works out children's doses by body weight (mg/kg).

Security and records

- Login is limited to official @moh.gov.my accounts.
- Access depends on the user's role: Administrator, Family Medicine Specialist, Medical Officer and Pharmacist.
- Every request, specialist approval, dispensing and AI use is recorded with the user's name, role and a timestamp.
- All patient data is stored in Malaysia, in line with the Personal Data Protection Act 2010.

7. What patients gain

1. Shorter waits at the pharmacy counter: under 7 minutes instead of 15 for a correct prescription, and 10 to 15 minutes instead of about 20 for a Controlled Drug.
2. No more trips back to the doctor's room. On average, 10 patients a month used to be sent back because of an incomplete AMS form. That no longer happens.
3. Safer prescriptions. The G6PD warning, the weight-based dose for children and the NAG 2024 dose suggestion are all checked when the prescription is written, before the medicine reaches the patient.
4. Antibiotics checked against the national guideline (NAG 2024) for every controlled antibiotic prescription.
5. Better privacy. Patients' clinical details are no longer passed around as WhatsApp screenshots.
6. More pharmacist time for patients. Time once spent searching WhatsApp, calculating quotas and typing into Excel can go to medication counselling.

8. Cost

The Innovation Team built PharmaSync in-house, so there was no development cost. Running it costs RM13 a year for the domain and RM100 a month for a server located in Malaysia, about RM3.32 a day, or RM1,213 a year. No new hardware is needed. Staff open it in the web browser already on the clinic computers.

AMS forms and quota lists are no longer printed. Before PharmaSync the pharmacy used one toner cartridge (RM800) every 3 months and one ream of A4 paper (RM12.90) every 2 months, which is RM3,277.40 a year. After the running cost, the clinic saves RM2,064.40 a year.

9. Use in other clinics

PharmaSync was built for more than one clinic. Each clinic's records are kept separate, while the UKK quota is shared at district level. The user roles match the existing staff structure of KKM health clinics,

so another clinic can adopt the system without rebuilding it or changing its organisation. Other health clinics under PKDJB now use PharmaSync, after it began as a trial at Klinik Kesihatan Kempas.

10. Innovation team: Kempas Innovators

Name	Position
Subramaniam A/L Palaniapan	Pharmacist UF12, Project Leader
Liew Bih Chung	Pharmacist UF14
Dr Amelia Teh Jay Yun	Family Medicine Specialist UD14
Dr Norlaila Najwa Binti Hasbullah	Medical Officer UD12
Ong Xiao Jing	Pharmacist UF9
Angel Lim	Pharmacist UF9

11. Conclusion

PharmaSync puts the whole Controlled Drug process in one system: the request, the quota check, the FMS decision and the dispensing record. The digital AMS form removes incomplete antibiotic forms, and the AI-assisted dose suggestion puts the NAG 2024 dose in front of the doctor while the prescription is written. Patients wait less, are no longer sent back to the doctor's room, and receive antibiotics whose dose has been suggested from the national guideline and checked for G6PD risk. It runs for about RM3.32 a day, saves RM2,064.40 a year in paper and toner, and is already used by other clinics in the district.

Source: PharmaSync Innovation Project Report, Pharmacy Unit, Klinik Kesihatan Kempas. Times are from Pharmacy Unit observation.